

KALAELOA, HI, USA

WMO: 911780

Lat: 21.317N

Lon: 158.067W

Elev: 10

StdP: 101.20

Time Zone: -10.00 (W10)

Period: 94-19

WBAN: 22551

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	17.0	10.7	8.0	19.8	11.8	8.6	20.4	10.3	24.2	9.0	24.6	1.6	40	0.311

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	32.1	23.4	31.3	23.1	31.0	23.0	25.5	29.4	24.9	29.0	24.4	28.7	4.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.1	27.7	23.5	18.4	27.3	22.9	17.6	26.9	77.9	29.4	75.2	29.0	73.2	28.8	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.2	6.3	DB	12.7	33.0	3.6	1.3	10.1	33.9	8.1	34.7	6.1	35.4	3.5	36.4	
			WB	11.3	26.4	1.7	1.7	10.1	27.7	9.1	28.7	8.2	29.6	7.0	30.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.8	22.5	22.6	23.0	23.9	24.7	25.9	26.7	27.0	26.7	25.9	24.7	23.4
	DBStd	2.00	1.49	1.38	1.31	1.27	1.25	0.99	0.97	0.89	1.09	1.24	1.24	1.35
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5384	388	352	402	418	455	478	517	526	499	493	440	415
	CDD18.3	2344	131	119	144	168	197	228	258	268	249	236	190	157
	CDH23.3	17482	634	585	747	1019	1443	1918	2364	2543	2276	1921	1224	808
	CDH26.7	4582	40	38	73	166	324	584	806	891	776	565	242	77
Wind	WSAvg	3.1	3.2	3.2	3.3	3.3	3.1	3.2	3.2	3.1	2.8	2.8	2.9	3.1
Precipitation	PrecAvg	471	83	52	39	37	21	7	6	6	13	54	53	73
	PrecMax	838	278	203	168	306	112	42	16	42	68	342	297	310
	PrecMin	110	4	0	1	1	0	1	0	0	2	1	0	2
	PrecStd	205	81	54	41	57	26	9	4	8	13	85	62	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.4	28.4	29.1	30.1	31.0	32.1	32.4	32.5	32.5	31.9	30.5	29.0
		MCWB	21.4	21.3	21.1	21.7	22.8	23.2	23.2	23.7	23.6	23.8	23.1	22.1
	2%	DB	27.6	27.6	28.0	29.1	29.9	31.1	31.6	31.9	31.7	31.0	29.5	28.0
		MCWB	21.3	21.1	20.9	21.4	22.1	22.6	23.1	23.4	23.3	23.4	22.8	21.7
	5%	DB	27.2	27.2	27.5	28.3	29.1	30.2	31.0	31.2	31.1	30.2	28.8	27.5
		MCWB	21.1	20.9	20.9	21.2	21.8	22.2	22.9	23.1	23.2	23.1	22.5	21.4
	10%	DB	26.4	26.4	27.0	27.6	28.4	29.5	30.2	30.6	30.2	29.4	28.0	27.1
		MCWB	20.8	20.5	20.7	20.9	21.7	22.0	22.6	22.9	23.0	22.9	22.2	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.2	23.9	23.8	24.1	24.5	24.7	25.6	26.5	25.8	26.1	25.6	24.4
		MCDB	26.6	26.7	26.3	27.6	28.9	29.6	29.9	30.2	29.8	29.4	28.6	27.0
	2%	WB	23.1	22.9	22.9	22.9	23.6	23.9	24.6	25.3	25.1	24.9	24.5	23.4
		MCDB	25.7	25.9	26.0	27.2	28.2	29.1	29.3	29.6	29.4	29.0	27.8	26.3
	5%	WB	22.5	22.2	22.2	22.3	23.0	23.2	24.0	24.5	24.5	24.3	23.8	22.8
		MCDB	25.5	25.6	25.9	26.8	27.6	28.5	29.0	29.2	29.0	28.6	27.2	26.0
	10%	WB	21.7	21.4	21.4	21.6	22.4	22.7	23.4	23.9	24.0	23.8	23.0	22.0
		MCDB	25.2	25.2	25.5	26.3	27.2	28.1	28.6	28.7	28.6	28.1	26.7	25.6
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.0	7.9	8.0	8.1	8.1	8.1	8.1	8.1	7.8	7.4	7.5
		MCDBR	8.3	8.4	8.4	8.4	8.5	8.5	8.5	8.5	8.5	8.3	7.8	7.8
		MCWBR	4.4	4.3	4.1	3.7	3.7	3.3	3.2	3.3	3.3	3.5	3.7	3.8
	5% WB	MCDBR	7.1	7.3	7.7	8.0	8.0	8.1	7.9	7.9	7.8	7.7	7.2	7.0
		MCWBR	4.4	4.4	4.5	4.3	4.2	3.9	3.8	3.9	3.9	4.0	4.3	4.1
Clear-Sky Solar Irradiance	taub	0.358	0.365	0.373	0.390	0.383	0.371	0.369	0.368	0.361	0.373	0.371	0.354	
	taud	2.591	2.562	2.548	2.481	2.532	2.573	2.589	2.591	2.615	2.592	2.589	2.625	
	Ebn at Noon	917	930	932	913	910	913	915	922	930	910	896	910	
	Edh at Noon	91	99	104	113	106	101	99	99	96	94	90	85	
All-Sky Solar Radiation	RadAvg	4.47	5.03	5.76	6.50	6.95	7.25	7.15	6.87	6.42	5.41	4.57	4.13	
	RadStd	0.19	0.36	0.61	0.37	0.40	0.21	0.15	0.28	0.19	0.18	0.12	0.30	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
										+68	+68

Nomenclature: See separate page